

Department of Leather Engineering
Khulna University of Engineering & Technology
B. Sc. Engineering 3rd Year 1st Term Examination-2019
Leather Manufacturing Technology IV
LE 3101

Time: 3 Hours.

Full Marks: 210

- N.B. i) Answer any THREE questions from each section in separate scripts.
 ii) Figures in the right margin indicate full marks.
 iii) Assume reasonable data if any missing.

SECTION-A

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|------|---|----|
| 1(a) | Discuss constant drying rate period in leather drying. | 07 |
| 1(b) | Write down the advantages and limitations of the overhead drying and toggling. | 10 |
| 1(c) | State any three common drying defects with their possible effects and remedies. | 09 |
| 1(d) | List out the key characteristics of glove, suede and glaze-kid leather. | 09 |
| 2(a) | State the principle of waterproofing of leather. | 09 |
| 2(b) | Assume that you have been given wet blue cow hide leather shaved to 1.4 to 1.5 mm. Formulate a recipe for the production of water proof shoe upper crust from it. | 10 |
| 2(c) | What is Nappa leather? Mention the general properties of nappa leather. | 07 |
| 2(d) | Write the controlling measures to be taken in the sub process of post tanning operation for an ideal nappa leather production. | 09 |
| 3(a) | State the principles of glaze-kid leather production. | 07 |
| 3(b) | What should be the physical properties of glove leather? Discuss the criteria need to consider during the selection of raw material for making glove leather? | 08 |
| 3(c) | Generally the key factors of glove leather largely achieved through chrome tanning but you are said to produce glove leather from vegetable tanned sheep skin leather. Now design the subsequent wet-end processes for imparting as much as possible the ideal features of glove leather. | 12 |
| 3(d) | What should be your choice of tanning method for book binding leather manufacturing and why? | 08 |
| 4(a) | How could you impart desired properties to the picking band leather through controlling the tanning operation? | 09 |
| 4(b) | Write the controlling techniques to the lining and bating operation for the manufacturing of lining leather. | 08 |
| 4(c) | Write down the selection of raw materials for the production of following leathers:
(i) Suede (ii) Nappa and (iii) Shoe upper | 08 |
| 4(d) | Why wet buffing is recommended for suede leather production? | 04 |
| 4(e) | Formulate a recipe for full shade dyeing of suede leather. | 06 |

SECTION-B

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|------|--|----|
| 5(a) | Write about the preparation of leather for finishing. | 04 |
| 5(b) | Define conditioning. Why conditioning is done before finishing? | 06 |
| 5(c) | Write short notes on the terms: (i) Staking (ii) Milling (iii) Snuffing and (iv) Polishing. | 12 |
| 5(d) | Define buffing. Discuss the function of buffing. Point out the impact of improper buffing of leather. | 09 |
| 5(e) | Define dedusting. Why dedusting is necessary before finishing? | 04 |
| 6(a) | Define finishing. What are the purposes of finishing? | 08 |
| 6(b) | Write the name of different layers of finish film with neat sketch. | 05 |
| 6(c) | Mention the classification of finishes according to finishing techniques, effects and materials. | 09 |
| 6(d) | Write the objectives of top coating. What should be the criteria of finishing for top coating of leather? | 07 |
| 6(e) | Discuss the terms: (i) Adhesion coat (ii) Pull up effects and (iii) Stucco | 06 |
| 7(a) | Define lacquer. Why lacquer is used in leather finishing? | 06 |
| 7(b) | Write about NC lacquer used in leather finishing with its advantages and disadvantages. | 06 |
| 7(c) | What is resin? Discuss the role of resin in leather finishing. | 06 |
| 7(d) | Differentiate between acrylic and polyurethane binder. | 05 |
| 7(e) | Discuss the terms: (i) Anti-sticking agents (ii) Filler (iii) Modifiers and (iv) Penetrators. | 12 |
| 8(a) | If an end product requires excellent flexibility and cold resistivity with strong covering properties which binder(s) will you choose and why? | 08 |
| 8(b) | List out the features of Roller coating machine. | 07 |
| 8(c) | Briefly discuss the following finishing defects, its causes and remedies: (i) Spray streaks (ii) Marking off (iii) Bronzing (iv) Orange peel effect and (v) Migration. | 15 |

Department of Leather Engineering
Khulna University of Engineering & Technology
B. Sc. Engineering 3rd Year 1st Term Examination-2019
Analytical Chemistry for Leather Manufacture
LE 3115

Time: 3 Hours.

Full Marks: 210

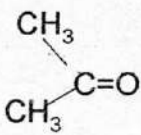
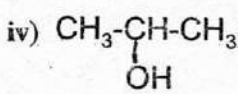
- N.B. i) Answer any THREE questions from each section in separate scripts.
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 iii) Assume reasonable data if any missing.

SECTION-A

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|------|--|----|
| 1(a) | Define accuracy, precision and significant figure with examples. | 06 |
| 1(b) | Write down the salient features of the analytical techniques:
i) Electroanalytical ii) Spectroscopy iii) Chromatography and iv) Titrimetry. | 08 |
| 1(c) | Classify analytical method based on sample size. | 05 |
| 1(d) | Define errors. Write down the steps for minimization of errors. | 04 |
| 1(e) | Define the terms: i) Calibration curve ii) Limit of detection and iii) limit of quantification.
If the blank responses of an analytical instrument are 0.011, 0.015, 0.030, 0.059, 0.074 and 0.089 then calculate the standard deviation (STDEV), limit of detection (LOD) and limit of quantification (LOQ) of that analytical instrument. | 12 |
| 2(a) | Define gravimetric analysis, precipitate and precipitant. Write down the properties of the precipitate and precipitating agents. | 09 |
| 2(b) | Define phase transition. Describe phase transition with a neat sketch. | 06 |
| 2(c) | Briefly describe a method for chromium determination in the spent chrome liquor other than instrumental method. | 08 |
| 2(d) | Define thermal decomposition curve. Briefly discuss the instrumentation of thermogravimetric analysis. | 12 |
| 3(a) | Draw the thermogram if you are provided copper sulphate ($\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$) where the temperature was increased ambient to above 500°C . The system was made inert by purging the N_2 gas. | 05 |
| 3(b) | Define chromatography and ion chromatography. Write down the application of ion chromatography in leather processing. | 06 |
| 3(c) | Briefly discuss the parameters that affect the qualities of an eluent. | 12 |
| 3(d) | Write short notes on analytical column and retention time. | 08 |
| 3(e) | The calibration curve for the standard chloride (Cl^-) ion (mg/L) by ion chromatography was obtained $y = 0.0023x + 0.0213$, pickle liquor showed the response at the same retention time 3.3192 units. Calculate the chloride concentration of the pickle liquor. | 04 |
| 4(a) | Briefly discuss the ion separation mechanism of ion chromatography. | 05 |
| 4(b) | Write down the principle of gas chromatography. | 04 |
| 4(c) | Write down the role of carrier gas in gas chromatography. Briefly state the selection criteria for carrier gas in terms of gas chromatography. | 09 |
| 4(d) | Distinguish between gas chromatography (GC) and high-performance liquid chromatography (HPLC). | 05 |
| 4(e) | Outline the instrument of gas chromatography. Briefly discuss the following terms in case of gas chromatography: i) Column and ii) Detector. | 12 |

SECTION-B

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|------|--|----|
| 5(a) | Define the terms: i) Eluent ii) Eluate iii) Mobile phase and iv) Chromatogram. | 08 |
| 5(b) | Discuss the basic principle of HPLC. Draw a schematic diagram of HPLC. | 08 |
| 5(c) | Mention the main function of following components of HPLC: i) Inlet filter ii) Solvent degassing system iii) Pump iv) Sample injection system v) Analytical column and vi) Guard column. | 09 |
| 5(d) | Distinguish between normal phase and reverse phase HPLC. | 05 |
| 5(e) | Write down the application of HPLC in different areas including leather science. | 05 |
| 6(a) | Define spectroscopy. Briefly discuss the theory of UV-Vis spectroscopy. | 09 |
| 6(b) | Deduce the mathematical expression of Beer-Lambert law with limitations. | 09 |
| 6(c) | Discuss briefly the function of different components of UV-Vis spectroscopy with its schematic diagram. | 08 |
| 6(d) | Distinguish between single and double beam UV spectrometer. | 05 |
| 6(e) | If the absorbance of zinc (Zn) in the spectrophotometer is 0.435, path length is 3.25 cm and molar concentration of Zn is 0.731. Calculate the molar absorption coefficient. | 04 |

- 7(a) What is the fingerprint region in IR spectroscopy? Why it is important in IR spectroscopy to analyze sample? 04
- 7(b) Illustrate the different types of vibrations in IR with a proper sketch. 09
- 7(c) Calculate the vibrational modes of the followings molecules: i) NH_3 ii) BeH_2 and iii) N_2O_4 . 03
- 7(d) Define NMR spectroscopy. Discuss the basic principle of NMR spectroscopy with an appropriate figure. 08
- 7(e) Calculate the NMR signals of the following compounds: 04
- i)  ii) $\text{CH}_3\text{-CH}_2\text{-NH}_2$ iii) $\text{CH}_3\text{-OH}$ iv) 
- 7(f) Define shielding and chemical shift. Describe n+1 rule for splitting of NMR signals in a compound with proper example. 07
- 8(a) Define mass spectroscopy. Write the basic principle of mass spectroscopy. 06
- 8(b) What does a mass analyzer do in MS? Describe ToF and Quadrupole mass analyzer shortly. 08
- 8(c) Draw a schematic diagram of AAS and describe hollow cathode lamp radiation source of AAS. 08
- 8(d) Describe the function and types of atomizer in AAS. Illustrate how the sample gets atomized in AAS. 09
- 8(e) Define ICP-MS. Why it is better than AAS? 04

Department of Leather Engineering
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B. Sc. Engineering 3rd Year 1st Term Examination-2019
Leather Products Engineering I
LE 3109

Time: 3 Hours. Full Marks: 210

N.B. i) Answer any THREE questions from each section in separate scripts.
 ii) Figures in the right margin indicate full marks.
 iii) Assume reasonable data if any missing.

SECTION-A

- | | |
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| 1(a) Define fashion and style. Why does fashion change frequently in our society-Explain? | 10 |
| 1(b) Write down the basic aspects of fashion. | 06 |
| 1(c) Distinguish between underlay allowances and overlay allowances. | 07 |
| 1(d) Write down the name of the categories of fashion and explain the characteristics of the following categories of fashion: (i) Elegant (ii) Classic and (iii) Western | 12 |
| 2(a) What is clicking? State the qualities are required for a perfect clicker. | 08 |
| 2(b) Define pattern allowance and 90° pierce. | 10 |
| 2(c) Differentiate between leather goods and leather products. | 05 |
| 2(d) Write down the classification of leather products according to volume and degree of reinforcement with example. | 12 |
| 3(a) What is perspecting drawing? Sketch the perspective view of a Bi-fold wallet. | 07 |
| 3(b) Write down the norms of pattern placement for leather products. | 08 |
| 3(c). Calculate the leather consumption of a complete Bi-fold wallet. | 12 |
| 3(d) Describe the following operations during fabrication of a wallet:
(i) Cradle construction and (ii) Asther preparation. | 08 |
| 4(a) Cite the points to be remembered during cutting operations. | 07 |
| 4(b) Why the interlocking system is important for cutting operation? | 05 |
| 4(c) Mention the concerning points for grading of leather. | 06 |
| 4(d) Why allowance is important for leather goods manufacturing? | 05 |
| 4(e) Explain- Factors should be considered during material consumption in leather products manufacturing. | 12 |

SECTION-B

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|--|----|
| 5(a) What is meant of the blueprint of a product? Classify pattern according to garments. | 08 |
| 5(b) Describe the techniques of designing a school bag with necessary sketches. | 12 |
| 5(c) List of the techniques of grading. Show the grading of a garment block through the template method. | 10 |
| 5(d) How does enlargement differ from grading? | 05 |
| 6(a) Point out the specifying parameters of colour. | 06 |
| 6(b) How do you apply the single split complementary colour scheme in designing leather clothing? | 07 |
| 6(c) Describe the basic colour scheme. | 14 |
| 6(d) Clarify the term-“Subjective nature of colour vision”. What is the colour wheel? | 08 |
| 7(a) List out the principle of design. | 05 |
| 7(b) Write down the applications of texture and line in footwear designing. | 07 |
| 7(c) Write down the functions of the following sewing machine parts:
(i) Pressure foot (ii) Feed dog (iii) Stich regulator and (iv) Reverse feed lever. | 16 |
| 7(d) What is run in operation? Mention its necessity. | 07 |
| 8(a) List out the operations in manufacturing a ladies shoulder bag. | 05 |
| 8(b) Clarify the terms- (i) Sewing (ii) Stitch and (iii) Seam | 09 |
| 8(c) Show the following pattern constructions with suitable sketches:
(i) Front and back (ii) Bottom and (iii) Gusset | 15 |
| 8(d) What are notches? Why it is necessary? | 06 |

Department of Leather Engineering
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B. Sc. Engineering 3rd Year 1st Term Examination-2019
Dyeing and Finishing I
LE 3107

Time: 3 Hours.

Full Marks: 210

- N.B. i) Answer any THREE questions from each section in separate scripts.
 ii) Figures in the right margin indicate full marks.
 iii) Assume reasonable data if any missing.

SECTION-A

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|---|----|
| 1(a) Explain bathochromic and hypsochromic effect. State the groups which influence these effects. | 09 |
| 1(b) How absorption radiation influences in color production. | 09 |
| 1(c) What is color index (C. I.)? State the importance of color index in leather dyeing. | 06 |
| 1(d) Explain briefly the pigmentary color production with example. | 06 |
| 1(e) Differentiate between additive and subtractive color mixing. | 05 |
| 2(a) What is the difference between chemiluminescence and fluorescence? | 06 |
| 2(b) What is Bucherer reaction? How will you get naphthoic acid by Bucherer reaction? | 06 |
| 2(c) Write the industrial route of "H-acid" preparation from naphthalene and prepare a dye from this acid. | 10 |
| 2(d) How phenol is prepared through (umene process)? | 05 |
| 2(e) What is fluorescence? Write down the principle of fluorescence. | 08 |
| 3(a) Define the following terms: i) Hue ii) Saturation iii) Brilliance. | 06 |
| 3(b) What are the differences between acid dye and basic dye? | 05 |
| 3(c) Explain the nomenclature of azo dye. Give an example of $A^1 \rightarrow Z \leftarrow A^2$ type azo dye. | 09 |
| 3(d) Define metal complex dye (MCD). Discuss the mechanism of MCD with leather fibers. | 10 |
| 3(e) Briefly discuss the factors influencing leather dyeing. | 05 |
| 4(a) Define nitration. How will you prepare p-nitro aniline from acetanilide by nitration reaction? | 06 |
| 4(b) State few tips regarding color matching during dyeing of crust leather. | 07 |
| 4(c) Explain the role of levelling agents and iso-electric point in dyeing. | 12 |
| 4(d) Explain the application of anionic auxiliary in drum dyeing with corresponding recipe. | 10 |

SECTION-B

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|---|----|
| 5(a) What is meant by surface protection and constructional ability in leather finishing? | 06 |
| 5(b) Define curtain coating. State briefly the ideal features of leather prior to finishing operations. | 09 |
| 5(c) Suppose you have supplied a D grade crust with light blue color from cow origin, make a short design of the construction phases of finishing for the production of light brown opaque finished leather. | 20 |
| 6(a) What is function of refractive index in film formation? Differentiate between organic and inorganic pigment. | 09 |
| 6(b) What is hybrid resin? Discuss the pros and cons of hybrid resin. | 08 |
| 6(c) During or after finish of leather surface showing behavior of i) finish sticks to plate ii) uneven sparying and iii) print cutting through the finish; as an engineer how can you fix and control this behavior? | 12 |
| 6(d) Why plasticizers are used in leather finishing? | 06 |
| 7(a) What is function of impregnation coat? Describe shortly the classification of leather finish based on finishing effect. | 10 |
| 7(b) Write a short note on the following finishing materials: i) Penetrator ii) Nitrocellulose lacquer and iii) Polyamide binder. | 12 |
| 7(c) Briefly describe the preparation of polyurethane binder and state the properties imparted to leather. | 13 |
| 8(a) State the theory of film formation and adhesion of finish. | 14 |
| 8(b) Why contact angle is essential in finish adhesion? Make a short description on influencing factors of finish adhesion. | 11 |
| 8(c) Define coating. Briefly state the physico-chemical requirements of the finishing coats that it should satisfy. | 10 |